

Low Level Design (LLD) — Online Medical Consultation Platform

1. Authentication Module LLD

APIs

POST /auth/login

Authenticates users.

POST /auth/register

Registers new users.

POST /auth/verify-token

Validates JWT tokens.

Authentication Flow

User Login



Firebase Authentication



JWT Generation



Session Validation



Role Verification

2. Appointment Module LLD

APIs

API	Purpose
POST /appointments/book	Create booking
GET /appointments/:id	Fetch appointment
PUT /appointments/cancel	Cancel booking

Booking Logic

1. Validate slot
2. Start Firestore transaction
3. Create lock
4. Process payment
5. Confirm booking

6. Generate meeting room
7. Send notifications

3. Doctor Assignment Engine LLD

Algorithm

Round Robin Allocation

Input:

- Available doctors
- Active appointment count

Process:

1. Remove unavailable doctors
2. Remove slot conflicts
3. Sort by active load
4. Apply round robin pointer
5. Select doctor

4. Slot Locking Engine LLD

Lock Structure

```
{  
  "slot_id": "2026-07-01-10:00",  
  "doctor_id": "DOC001",  
  "locked_by": "USER001",  
  "created_at": "timestamp",  
  "expires_at": "timestamp"  
}
```

Lock Flow

Check Existing Lock



Lock Exists?



Validate Expiry



Override Stale Lock



Create New Lock

5. Payment Module LLD

Payment Flow

Create Razorpay Order
↓
Initialize Checkout
↓
Payment Callback
↓
Signature Verification
↓
Booking Confirmation

APIs

API	Purpose
POST /payment/create-order	Create payment order
POST /payment/verify	Verify payment

6. Video Consultation Module LLD

Meeting Flow

Appointment Confirmed
↓
Generate Jitsi Room
↓
Share Meeting URL
↓
Join Session
↓
Realtime Chat
↓
End Session

7. Notification Service LLD

Notification Types

- Email Notifications
 - Push Notifications
 - Reminder Notifications
 - Dashboard Alerts
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Reminder Scheduler

Reminder	Time
Reminder 1	24 Hours Before
Reminder 2	1 Hour Before
Reminder 3	10 Minutes Before

8. Firestore Collections LLD

users

```
{
  "user_id": "USR001",
  "name": "Robin",
  "email": "example@gmail.com",
  "role": "PATIENT"
}
```

doctors

```
{
  "doctor_id": "DOC001",
  "specialization": "Cardiology",
  "status": "ACTIVE",
  "available": true
}
```

appointments

```
{
  "appointment_id": "APT001",
  "patient_id": "USR001",
  "doctor_id": "DOC001",
  "slot": "10:00 AM",
  "status": "BOOKED"
}
```

9. State Management LLD

Appointment States

PENDING
LOCKED
PAYMENT_PROCESSING
BOOKED

CONSULTATION_ACTIVE
COMPLETED
CANCELLED
PAYMENT_FAILED
EXPIRED

10. Error Handling Strategy

Error Type	Handling
Payment Failure	Release lock
Doctor Unavailable	Reassign doctor
Session Expired	Force re-login
Network Failure	Retry mechanism
Lock Conflict	Reject booking

11. Logging & Monitoring

Monitoring Tools

- Cloud Logging
 - Firebase Monitoring
 - API Metrics
 - Transaction Tracking
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12. Future Enhancements

- AI-based doctor recommendation
 - E-prescription generation
 - Voice consultation support
 - Multi-language support
 - Analytics dashboard
 - Appointment prediction engine
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End of LLD Documentation